

2(a)	any two from: 1 (initial) volume / population / number, of bacteria (in bacterial suspensions) or volume of (bacterial) suspension or (original) optical density of (bacterial) suspensions ; 2 volume of nitrite (ion solution) ; 3 concentration / volume, of nutrient broth ;	2
2(b)(i)	(<i>species of bacterium</i> :) nominal ; (<i>lowest nitrite concentration</i> :) continuous ;	2
2(b)(ii)	1 correct reading of values from Figure 2.1 (<i>Salmonella enteritidis</i> =) 0.7 <u>and</u> (<i>Salmonella typhimurium</i> =) 5 ; 2 correct working ; 3 correct value for percentage difference ; 614 (%) / 86 (%)	3
2(b)(iii)	(most affected =) <i>Yersinia enterocolitica</i> (least affected =) <i>Salmonella typhimurium</i> ;	1

Question	Answer	Marks
2(c)	<i>any four from:</i> 1 <i>idea that</i> investigation, not carried out in a person / carried out in a laboratory ; 2 <i>idea that</i> investigation only carried out for 24 hours / no information about longer term effects ; 3 <i>idea that</i> experiment, only used 5 species of (pathogenic) bacteria / did not use other bacteria ; 4 pH in the, digestive system / stomach / intestines / AW, is not 4.8 / is different ; 5 <i>idea that</i> nitrites may, kill / reduce growth / AW, of / beneficial / AW, bacteria (in the digestive system) ; 6 nitrites could, cause side effects / be toxic / AW (in humans) ; 7 no evidence of control experiment ; 8 no statistical, test / analysis ; 9 <i>idea that</i> (bacterial) population sizes may not have decreased (as only measured when no bacterial population growth occurred) ;	4